

Displays

D1:

$$x(k+1) = Ax(k) + Bu(k) + \bar{w}(k) + w(k) \quad (1)$$

D2:

$$\Pr(x(k) \in \mathcal{X}^j \mid x(0)) \geq p_j, \quad j \in \{1, \dots, n_c\}, \quad (2a)$$

D3:

$$u(k) \in \mathcal{U}, \quad (2b)$$

D4:

$$z(k+1) = Az(k) + Bv(k) + \bar{w}(k), \quad (3a)$$

D5:

$$e(k+1) = Ae(k) + B\pi_{\text{tube}}(e(k)) + w(k) \quad (3b)$$

D6:

$$\Pr(e(k) \in \mathcal{R}_k \mid e(0)) \geq p.$$

D7:

$$\Pr(e(k) \in \mathcal{R} \mid e(0)) \geq p \quad \forall 0 \leq k \leq \bar{N}.$$

D8:

$$\min_{x \in \mathcal{X} \subseteq \mathbb{R}^d} c^\top x \quad (4a)$$

D9:

$$\text{s.t.} \quad \Pr(x \in \mathcal{X}_\delta) \geq p, \quad (4b)$$

D10:

$$\min_{x \in \mathcal{X} \subseteq \mathbb{R}^d} c^\top x \quad (5a)$$

D11:

$$\text{s.t.} \quad x \in \mathcal{X}_{\delta^{(i)}}, \quad i \in \mathcal{I}_s, \quad (5b)$$

D12:

$$\binom{N_k + d - 1}{N_k} \sum_{i=0}^{N_k + d - 1} \binom{N_s}{i} (1-p)^i p^{N_s - i} \leq \beta. \quad (6)$$

D13:

$$N_k \leq (1-p)N_s - d + 1 - \sqrt{2(1-p)N_s \ln \left(\frac{((1-p)N_s)^{d-1}}{\beta} \right)} \quad (7)$$

D14:

$$N_s \geq \frac{2}{1-p} ((d-1) \ln(2) - \ln(\beta)), \quad (8)$$

D15:

$$x_{i+1} = Ax_i + Bu_i + \bar{w}_i + w_i, \quad (9a)$$

D16:

$$z_{i+1} = Az_i + Bv_i + \bar{w}_i, \quad (9b)$$

D17:

$$e_{i+1} = Ae_i + B\pi_{\text{tube}}(e_i) + w_i, \quad (9c)$$

D18:

$$\pi_{\text{tube}}(e) \in \mathcal{E}_u \subset \mathcal{U}, \quad \forall e \in \mathbb{R}^n.$$

D19:

$$z_i \in \mathcal{Z}_i = \bigcap_{j=1}^{n_c} \left(\mathcal{X}^j \ominus \mathcal{R}_{k+i}^j \right), \quad (10a)$$

D20:

$$v_i \in \mathcal{V} = \mathcal{U} \ominus \mathcal{E}_u. \quad (10b)$$

D21:

$$\mathbb{E}_{W_k} \left(l_f(x_N) + \sum_{i=0}^{N-1} l_{k+i}(x_i, u_i) \right).$$

D22:

$$\min_{\{v_i\}} \sum_{l=1}^{N_s^{\text{MPC}}} \left(l_f(x_N^{(l)}) + \sum_{i=0}^{N-1} l_{k+i}(x_i^{(l)}, u_i^{(l)}) \right) \quad (11a)$$

D23:

$$\text{s.t. } x_{i+1}^{(l)} = z_{i+1} + e_{i+1}^{(l)} \quad (11b)$$

D24:

$$u_i^{(l)} = v_i + \pi_{\text{tube}}(e_i^{(l)}) \quad (11c)$$

D25:

$$e_{i+1}^{(l)} = A e_i^{(l)} + B \pi_{\text{tube}}(e_i^{(l)}) + w_i^{(l)} \quad (11d)$$

D26:

$$z_{i+1} = A z_i + B v_i + \bar{w}_i \quad (11e)$$

D27:

$$v_i \in \mathcal{V}, \quad z_i \in \mathcal{Z}_i, \quad z_N \in \mathcal{Z}_f \quad (11f)$$

D28:

$$z_0 = z(k), \quad x_0^{(l)} = x(k), \quad e_0^{(l)} = e(k), \quad (11g)$$

D29:

$$u(k) = v_0^* + \pi_{\text{tube}}(e(k)). \quad (12)$$

D30:

$$z \in \mathcal{Z}_f \Rightarrow A z + B \pi_f(z) + \bar{w} \in \mathcal{Z}_f$$

D31:

$$\begin{aligned} \min_{\alpha > 0} \quad & \alpha \\ \text{s.t.} \quad & \Pr(e(k) \in \alpha \tilde{\mathcal{R}}) \geq p. \end{aligned}$$

D32:

$$\min_{\alpha > 0} \quad \alpha \quad (13a)$$

D33:

$$\text{s.t. } e_k^{(i)} \in \alpha \tilde{\mathcal{R}}, \quad i \in \mathcal{I}_s, \quad (13b)$$

D34:

$$\min_{b > 0} \quad \|b\|_1 \quad (14a)$$

D35:

$$\text{s.t. } H e_k^{(i)} \leq b, \quad i \in \mathcal{I}_s. \quad (14b)$$

D36:

$$\min_{P > 0, e_c} \quad -\log \det P \quad (15a)$$

D37:

$$\text{s.t. } (e_k^{(i)} - e_c)^\top P (e_k^{(i)} - e_c) \leq 1, \quad i \in \mathcal{I}_s, \quad (15b)$$

D38:

$$[|p|, |v|]^\top \leq [p_{\max}, v_{\max}]^\top = [1, 0.4]^\top, \quad (16)$$

D39:

$$\Pr(\theta(k) \geq -0.08) \geq 90\%, \quad (17a)$$

D40:

$$\Pr(\theta(k) \leq 0.08) \geq 90\%. \quad (17b)$$

D41:

$$x_k^{\text{ref}} = \begin{cases} [1, 0, 0, 0]^{\text{T}}, & k \leq 100 \\ [-1, 0, 0, 0]^{\text{T}}, & k > 100 \end{cases}$$

D42:

$$\begin{bmatrix} \cos \theta & l \\ m + M & Ml \cos \theta \end{bmatrix} \begin{bmatrix} \ddot{p} \\ \ddot{\theta} \end{bmatrix} = \begin{bmatrix} -g \sin \theta - d_M \frac{\dot{\theta}}{M} + \frac{w}{M} \cos \theta \\ u - d_p \dot{p} - d_M (\dot{p} + l \dot{\theta} \cos \theta) + Ml \dot{\theta}^2 \sin \theta + w \end{bmatrix}$$

Inline

I1: $x(k) \in \mathbb{R}^n$

I2: $u(k) \in \mathbb{R}^{n_u}$

I3: $\mathbb{R}^n$

I4: $\bar{w}(k) \in \bar{\mathcal{W}}$

I5: $w(k)$

I6: $\bar{N}$

I7: $W = [w(0)^{\text{T}}, \dots, w(\bar{N})^{\text{T}}]^{\text{T}} \sim \mathcal{Q}$

I8: n_c

I9: $\mathcal{X}^j \subseteq \mathbb{R}^n$

I10: $\mathcal{U} \subseteq \mathbb{R}^{n_u}$

I11: $x(0)$

I12: $l_k(x(k), u(k))$

I13: $\bar{N} \rightarrow \infty$

I14: $\mathcal{U}$

I15: $N \ll \bar{N}$

I16: $z(k)$

I17: $e(k)$

I18: $x(k) = z(k) + e(k)$

I19: $v(k)$

I20: π_{tube}

I21: $u(k) = v(k) + \pi_{\text{tube}}(e(k))$

I22: $z(0) = x(0)$

I23: $e(0) = 0$

I24: k

I25: $\mathcal{R}_k$

I26: $0 \leq k \leq \bar{N}$

I27: p

I28: $e(0)$

I29: $\mathcal{R}$

I30: W

I31: $\mathcal{X}_\delta$

I32: δ

I33: d

I34: x

I35: N_s

I36: $i \in \mathcal{I}_s$

I37: $\delta^{(i)}$

I38: $|\mathcal{I}_s| = N_s - N_k$

I39: N_k

I40: x^*

I41: $\mathcal{X}_{\delta^{(j)}}$

I42: $j \in \{1, \dots, N_s\} \setminus \mathcal{I}_s$

I43: $1 - \beta$

I44: β

I45: $N_k = 0$

I46: i

I47: $x_0 = x(k)$
 I48: $z_0 = z(k)$
 I49: $e_0 = e(k)$
 I50: $\bar{w}_i = \bar{w}(k + i)$
 I51: W_k
 I52: w_i
 I53: e_i
 I54: $W_k = [w_0, \dots, w_N]$
 I55: $p(W_k) = p\left([w(k)^\top, \dots, w(k + N)^\top]^\top \mid [w(0)^\top, \dots, w(k - 1)^\top]^\top\right)$
 I56: u
 I57: z
 I58: v
 I59: j
 I60: $\mathcal{R}_k^j$
 I61: $\mathcal{R}_{k+i}^j$
 I62: $k + i$
 I63: p_j
 I64: $\mathcal{Z}_f$
 I65: l_f
 I66: N_s^{MPC}
 I67: $i \in \{0, \dots, N - 1\}$
 I68: $e_i^{(l)}$
 I69: $\pi_{\text{tube}}(e_i^{(l)})$
 I70: $\{v_i\}$
 I71: $v(k) = v_0^*$
 I72: v_0^*
 I73: $x(k)$
 I74: $\bar{w} \in \bar{\mathcal{W}}$
 I75: $\pi_f(z) \in \mathcal{V} \ \forall z \in \mathcal{Z}_f$
 I76: $\mathcal{Z}_f \subseteq \mathcal{Z}_\infty$
 I77: $\mathcal{Z}_\infty = \bigcap_{k=1}^{\bar{N}} \mathcal{Z}_k$
 I78: $\bar{\mathcal{W}} = \{0\}$
 I79: $\mathcal{R}_k = \mathcal{R}$
 I80: $\mathcal{Z}_k$
 I81: $k = 0, \dots, \bar{N}$
 I82: $x(0) = z(0)$
 I83: $0 \leq k \leq \bar{N} - N$
 I84: $V^* = \{v_0^*, \dots, v_{N-1}^*\}$
 I85: $Z^* = \{z_0^*, \dots, z_N^*\}$
 I86: $x(k + 1)$
 I87: $z(k + 1) = z_1^*$
 I88: $\bar{V} = \{v_1^*, \dots, v_{N-1}^*, \pi_f(z_N^*)\}$
 I89: $\bar{Z} = \{z_1^*, \dots, z_N^*, Az_N^* + B\pi_f(z_N^*) + \bar{w}(k + N + 1)\}$
 I90: $v_i^* \in \mathcal{V}$
 I91: $1 \leq i \leq N$
 I92: $\pi_f(z_N^*) \in \mathcal{V}$
 I93: $\bar{V}$
 I94: $z_i^* \in \mathcal{Z}_i(k) = \mathcal{Z}_{i-1}(k + 1)$
 I95: $\mathcal{Z}_i(k)$
 I96: $Az_N^* + B\pi_f(z_N^*) + \bar{w}(k + N + 1) \in \mathcal{Z}_f$
 I97: $\square$
 I98: $u(k)$
 I99: $u(k) \in \mathcal{U}$
 I100: $\pi_{\text{tube}}(e) \in \mathcal{E}_u$
 I101: $e \in \mathbb{R}^n$

I102: $\Pr(e(k) \in \mathcal{R}_k^j) \geq p_j$
 I103: $z(k) \in \mathcal{Z}_0(k)$
 I104: $\mathcal{Z}_0(k) = \bigcap_{j=1}^{n_c} (\mathcal{X}^j \ominus \mathcal{R}_k^j)$
 I105: $\Pr(x(k) \in \mathcal{X}^j) \geq p_j$
 I106: $j = 1, \dots, n_c$
 I107: $W^{(i)} = [w_0^{(i)}, \dots, w_{N-1}^{(i)}]^\top \sim \mathcal{W}$
 I108: $i \in \{1, \dots, N_s\}$
 I109: $E^{(i)} = [e_0^{(i)}, \dots, e_N^{(i)}]^\top$
 I110: $e_0^{(i)} = e(0) = 0$
 I111: $e_k^{(i)} \in \mathcal{R}_k$
 I112: $\mathcal{X}^j$
 I113: $\tilde{\mathcal{R}}$
 I114: $\mathcal{R} = \alpha \tilde{\mathcal{R}}$
 I115: α
 I116: $\mathcal{X}_\delta := \{\alpha \mid e(k) \in \alpha \tilde{\mathcal{R}}\}$
 I117: $e_k^{(i)}$
 I118: $\mathcal{I}_s$
 I119: α^*
 I120: $d = 1$
 I121: $\alpha^* \tilde{\mathcal{R}}$
 I122: $\tilde{\mathcal{R}} = \{e \mid h^\top e \leq 1\}$
 I123: $h^\top e_k^{(i)}$
 I124: $\alpha^* = \max_{i \in \mathcal{I}_s} h^\top e_k^{(i)}$
 I125: $\tilde{\mathcal{R}} = \{e \mid He \leq \mathbf{1}\}$
 I126: $H \in \mathbb{R}^{n_{hs} \times n}$
 I127: $\mathbf{1} \in \mathbb{R}^{n_{hs}}$
 I128: $\|He_k^{(i)}\|_\infty$
 I129: H
 I130: b^*
 I131: $d = n_{hs}$
 I132: $\{e \mid He \leq b^*\}$
 I133: $d = \frac{n^2+n}{2} + n$
 I134: P
 I135: e_c
 I136: P^*
 I137: e_c^*
 I138: $\{e \mid (e - e_c^*)^\top P^{*-1}(e - e_c^*) \leq 1\}$
 I139: $e_c = 0$
 I140: $d = \frac{n^2+n}{2}$
 I141: $\mathcal{I}_{\text{dis}}$
 I142: $x = [p, v, \theta, r]^\top$
 I143: θ
 I144: r
 I145: $\bar{N} = 200$
 I146: w
 I147: $W \sim \mathcal{N}(0, \Sigma^w)$
 I148: $|u| \leq u_{\text{max}} = 4$
 I149: $x(0) = 0$
 I150: $(x_i - x_{k+i}^{\text{ref}})^\top Q(x_i - x_{k+i}^{\text{ref}}) + u^\top Ru$
 I151: $Q = I$
 I152: $R = 10^{-4}$
 I153: $N_s^{\text{MPC}} = 10$
 I154: $N = 30$
 I155: ± 0.4

I156: $N_s = 10000$
 I157: $\mathcal{R}_k^{|p|, |v|}$
 I158: $p_1 = 99.6\%$
 I159: $1 - \beta \approx 1 - 10^{-7}$
 I160: $k = 820$
 I161: $\mathcal{R}_k^{\theta_{\max}}$
 I162: $\mathcal{R}_k^{\theta_{\min}}$
 I163: $p_2 = 90\%$
 I164: $\beta = 10^{-7}$
 I165: $n_c = 3$
 I166: $p_{ref} = 1$
 I167: $k \leq 100$
 I168: $p_{ref} = -1$
 I169: $k > 100$
 I170: $\begin{bmatrix} |p| \\ |v| \end{bmatrix} \leq \begin{bmatrix} p_{\max} \\ v_{\max} \end{bmatrix}$
 I171: $\theta \leq \theta_{\max}$
 I172: $\theta \geq -\theta_{\max}$
 I173: $m = 1$
 I174: $d_m = 10$
 I175: $M = 1$
 I176: $d_M = 1$
 I177: $l = 1$
 I178: $g = 9.81$
 I179: $T_s = 0.1$
 I180: $\lambda = [1, 0.3672, 0.8617 \pm 0.2788i]^\top$
 I181: $W \sim \mathcal{N}(0, K)$
 I182: $K_{i,j} = 0.02^2 + 0.2^2 \exp(-\frac{1}{2}(i-j)^2/10^2)$
 I183: $i, j \in \{1, \dots, \bar{N}\}$
 I184: $\mathbf{T}$
 I185: $1 - \beta$
 I186: $\{1, \dots, n_c\}$
 I187: $\{v_0^*, \dots, v_{N-1}^*\}$
 I188: $\mathcal{Z}_\infty$
 I189: $k = 1$
 I190: $W^{(i)} \sim \mathcal{W}$
 I191: $W \sim \mathcal{Q}$
 I192: $W^{(i)}$
 I193: $\bar{N} - 1$
 I194: $E^{(i)}$
 I195: $P > 0$
 I196: P^{*-1}
 I197: p_{ref}
 I198: $\pm\theta_{\max}$
 I199: d_p